

Distinguished Lecture Review

Talk title	What we see and what we value: AI with a human perspective
Speaker	Prof. Fei-Fei Li (Stanford University)
Quarter/year talk was given	Winter 2024 (delivered Thursday, January 18, 2024)
Talk URL/link	https://www.youtube.com/watch?v=gzOwpEupP5w
Talk was a	Distinguished Lecture
Lecture notes	Lecture Notes
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1. Heart of the Talk & Great Ideas

Prof. Li's talk is organized around a simple, memorable question — *where have we come from, and where are we heading?* — and three theses that map onto it: **build AI to see what humans see; build AI to see what humans *don't* see; build AI to see what humans *want* to see.** I found this framing clarifying because it turns sixty years of computer-vision history into a human-centered arc rather than a list of architectures. It also lets the same talk reach back to the Cambrian Explosion and forward to household robots without feeling scattered.

The historical opening was the part I most enjoyed as a learner. Prof. Li opens with Andrew Parker's "Light Switch" conjecture — that the sudden evolution of **vision ~540 million years ago** triggered the Cambrian Explosion of animal species — and uses that to motivate why enabling machines to see is so consequential. The framing ("vision is what kicked off intelligence on this planet") gave the rest of the talk a through-line that I don't think I would have carried otherwise. I like how it re-situates computer vision as not just engineering, but continuing an evolutionary project.

Her three-phase telling of object recognition was the clearest version of this story I've come across. Phase 1 (hand-designed features + hand-designed models — Geons, Generalized Cylinders, Parts + Springs) she described as *theoretically beautiful, but they don't work*. Phase 2 (hand-designed features, learned models — Pictorial Structures, Bag of Words, CRFs, And-Or Graphs) added statistics but stayed narrow. Phase 3 (learned features *and* learned models) started with **ImageNet** (15M images / 22K categories) and **AlexNet** in 2012 — and the top-5 ImageNet error curve (28% → 16% with AlexNet → 3.6% with ResNet, crossing the human baseline around 2015) makes the jump visceral. Prof. Li's observation that **vision didn't just benefit from deep learning — vision contributed to it** re-centered ImageNet for me; it's easy to treat the dataset as scaffolding rather than catalyst.

The section I'll carry forward longest is **compositionality via scene graphs** (Visual Genome). Objects alone aren't enough to categorize a scene; relationships between objects — entities, attributes, inter-entity relations — must also be modeled. The zero-shot examples — *person riding horse + person wearing hat* → *horse wearing a hat, person sitting on chair + fire hydrant* → *person sitting on fire hydrant* — were a great way of showing how compositionality gives coverage over the combinatorial long tail without needing a labeled example of every weird combination. **MOMA** (Multi-Object Multi-Actor activity understanding) extended that compositional view from static scenes to activities over time, and I found that extension elegant.

The thesis-2 arguments on **fine-grained recognition** — species of birds, car make/model/year, Street-View-plus-2010-Census inferring income and voting patterns from neighborhood imagery — were the moments where I most strongly felt the “AI sees what humans don’t” framing. The priming-bias / surgical-instrument / hospital-acquired-infection examples under ambient intelligence were a good reminder that filling blind spots in human attention is a legitimate and maybe under-valued use of the technology.

Running through it all was the strongest unifying observation of the hour: that **data + compute + neural nets/algorithms** ushered in the deep-learning revolution, and that AI’s development has been — and will continue to be — inspired by brain science and human cognition. Prof. Li’s H-A-I framing for Stanford’s Human-Centered AI Institute — **Human impact, Augment** (don’t replace), **Inspired** by human intelligence — gave the talk a clean ethical closing.

2. What Could Be Improved

These are personal “as a learner” notes, not critiques of the underlying work.

Thesis 2 — “seeing what humans don’t see” — felt narrower than the framing invited. The Street-View-plus-2010-Census example was wonderful, but a few more would have helped. There’s a “telescope vs microscope” distinction I kept reaching for: humans individually see the world like a microscope; what would a *telescope* view look like for energy usage, natural-resource distribution, population migration, or emergency preparedness? Wildlife habitat mapping from social-media photos is another example in the same spirit. Relatedly — **other species see in ways humans can’t** (bees, mantis shrimp, snakes with infrared pits, plants responding to polarized light) and there’s real room to discuss applications or techniques inspired by non-human vision.

Thesis 3 — “seeing what humans want” — leaned heavily on healthcare. Healthcare is probably the largest problem space for human-centered AI, so the emphasis is fair. But safety in industrial settings, manufacturing-defect detection, and fatigue/inattention monitoring all have direct downstream effects on healthcare too — those pre-healthcare domains seemed under-represented relative to the breadth the framing promised. The later half of this section also drifted from vision into robotics — BEHAVIOR, OmniGibson, VoxPoser — which was interesting but meant less of the hour sat with “vision” as the lens.

Privacy — the PrivHAR framing was intriguing, but the auditability question felt open. Getting the signals a computer-vision system needs without revealing what humans see is a wonderful design stance. But it raises a follow-up: how is the warped-signal pipeline audited? How is a privacy-preserving pipeline traced to ensure it’s not silently degraded or rolled back? A few words on mathematical guarantees (differential-privacy-style bounds, signal-reconstruction impossibility results) or process/tooling would have strengthened the point.

The closing observations read more aspirational than mechanistic. The earlier parts of the talk — three phases of object recognition, the ImageNet arc, scene-graph compositionality — were mechanistic: here’s how the field evolved, here’s what changed, here’s why. The HAI closing — “AI must augment, not replace,” “multi-disciplinary approach” — I agreed with, but the *how* wasn’t as concrete as the earlier “*how*.” Directional techniques, incentives, or institutional mechanisms would have landed more strongly for me.

The 1000-household-activities list could be more inclusive. The sampled activities — “clean after wild party,” “clean a shower,” “scrub bathroom floor,” “shovel snow,” “play squash” — are clearly anchored in a particular demographic. What does the equivalent list look like in agricultural settings, or for dangerous professions (high temperature, high noise, vibration)? The payoff for automation in those settings is arguably much higher. Different

countries and societies have different tasks; BEHAVIOR is explicitly framed as “originated from human needs,” and which humans’ needs you start from really matters.

The behavioral question around pervasive cameras wasn’t addressed. Independent of how much data is collected, how does human behavior change when multiple cameras are installed around people? What’s the social impact of the *presence* of AI vision in a home or hospital? These questions seem squarely within the scope of human-centered AI and would have been a natural companion to the privacy section.

3. Conclusions / Key Takeaways

- **Vision history, lucidly told.** The three-phase framing of object recognition (hand-designed features and models → hand-designed features, learned models → learned features *and* learned models) is one of the clearest on-ramps to the deep-learning revolution I’ve come across.
- **Data and benchmarking as engines of innovation.** ImageNet, BEHAVIOR, and the repeated pattern of “define the benchmark → innovation follows” was the methodological point I most want to carry forward from the hour.
- **Compositionality and scene graphs.** Representing a scene as entities + attributes + relations is how you get zero-shot coverage over the combinatorial long tail. One of the more quietly beautiful ideas in the talk.
- **MOMA — static to dynamic relations.** Extending compositional structure from a single frame to an activity over time felt like a natural and productive generalization.
- **Fine-grained recognition as a real complement to human attention.** Bird species, car make/model/year, Street-View-to-demographics inference — these sit in the “AI sees what humans don’t” niche more comfortably than the more-hyped use cases.
- **The Cambrian framing.** Vision didn’t just co-evolve with intelligence; on Prof. Li’s telling it was the trigger. A frame I expect to keep paying off long after the hour ends.

Overall, one of the more enjoyable talks I’ve watched this quarter — the historical arc, the compositionality idea, and the H-A-I ethical frame are the threads I’m likely to return to.